

Chapter 8: Data Analysis, 2016 & 2017 data, Total Number x Total Area

Combining the 2017 data with the 2016 data filtered at 65. (This may not be correct)

Import Libraries

```
In [1]: import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib as mpl
from matplotlib import pyplot as plt

%matplotlib inline
```

Import the data

```
In [2]: totalsXX = None
totalsXX = pd.read_csv('TXX_Totals.csv')

totals65 = None
totals65 = pd.read_csv('T65_Totals.csv')
```

Combine data into a single dataframe

```
In [3]: frames = [totalsXX, totals65]
combined65 = pd.concat(frames)
```

Update 'Image Name' values for 2017 data

Assumes that the image is a mutant and that the 'M' value corresponds to the Mutant name

```
In [4]: receptor = [('A3M17', 'M17'), ('B1M29', 'M29')]

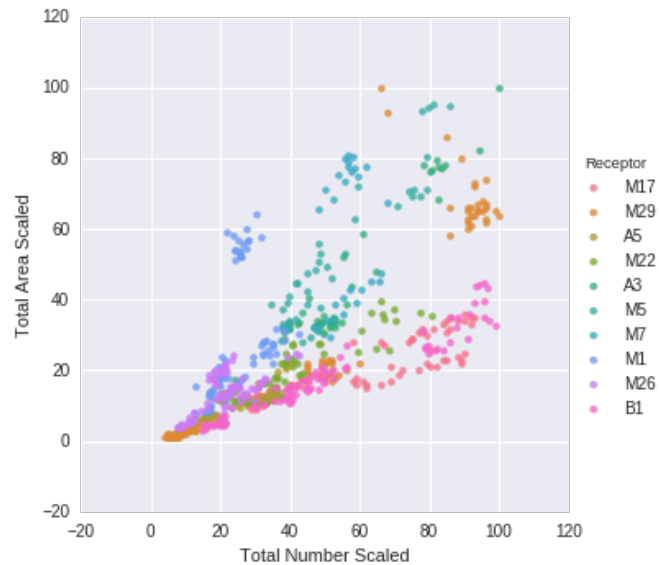
for r in receptor:
    combined65.loc[combined65['Image Name'].str.contains(r[0]), 'Receptor']
    = r[1]
```

Visualizations

Unless otherwise noted, the graphs will use the scaled values.

Number by area, both receptors, post steps, both runs

```
In [5]: number_by_area_all = sns.lmplot('Total Number Scaled', 'Total Area Scaled',  
                                         data=combined65,  
                                         fit_reg=False,  
                                         hue='Receptor')  
  
number_by_area_all.savefig('comb65_number_by_area_both.png')
```



Number by area, A3, post steps, scatter plot

```

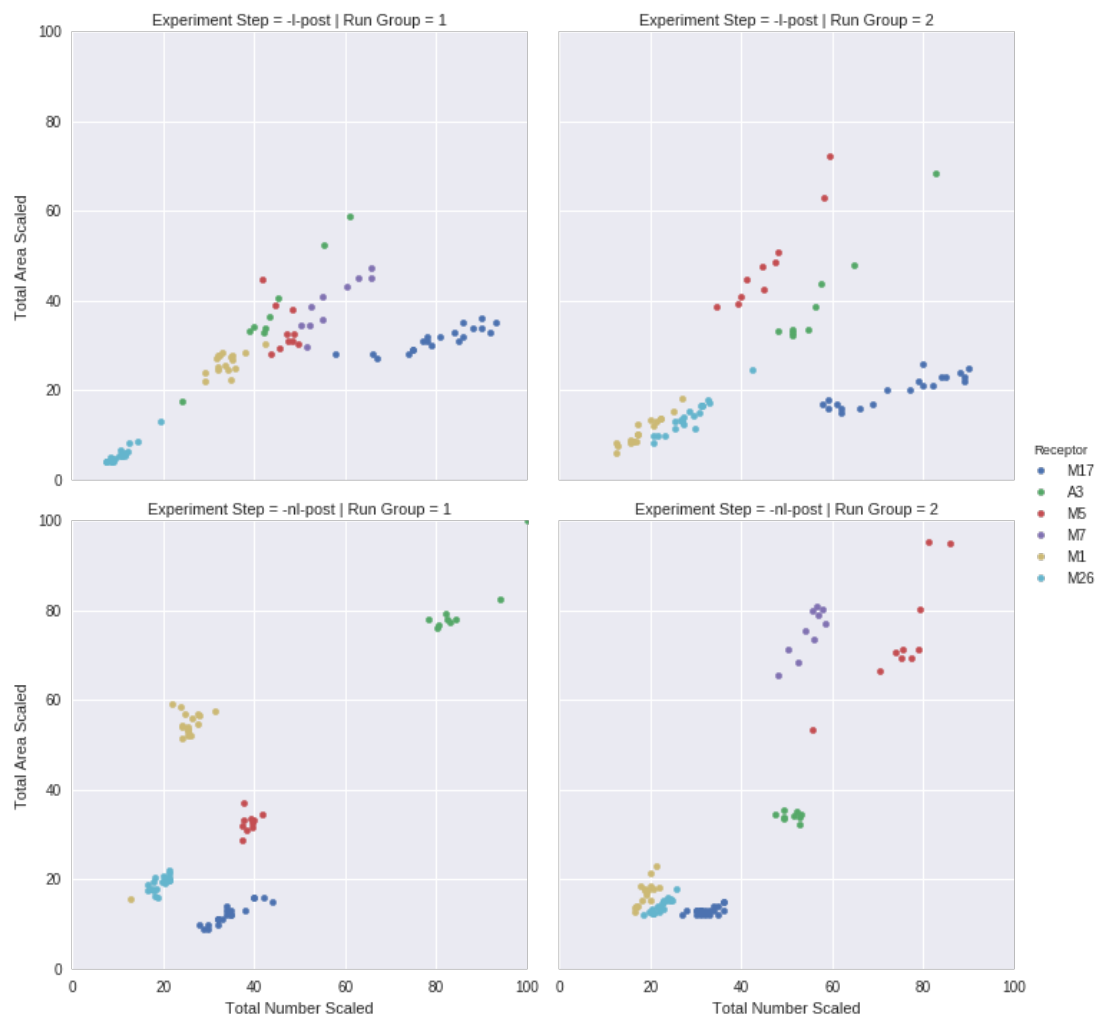
In [6]: post = ['-l-post', '-nl-post']
receptors = ['A3', 'M1', 'M5', 'M7', 'M17', 'M26']

tmpdf = None
tmpdf = pd.DataFrame(combined65[(combined65['Receptor'].isin(receptors)) &
                                (combined65['Experiment Step'].isin(post))
                                ])

g = sns.FacetGrid(tmpdf,
                  col='Run Group',
                  row='Experiment Step',
                  hue='Receptor',
                  size=5,
                  ylim=(0,100), xlim=(0,100),
                  aspect=1
                  )

g.map(plt.scatter, 'Total Number Scaled', 'Total Area Scaled')
g.add_legend()
g.savefig('comb65_num-by-area_a3_post-steps_scatter.png')

```



Number by area, B1, post steps, scatter plot

```

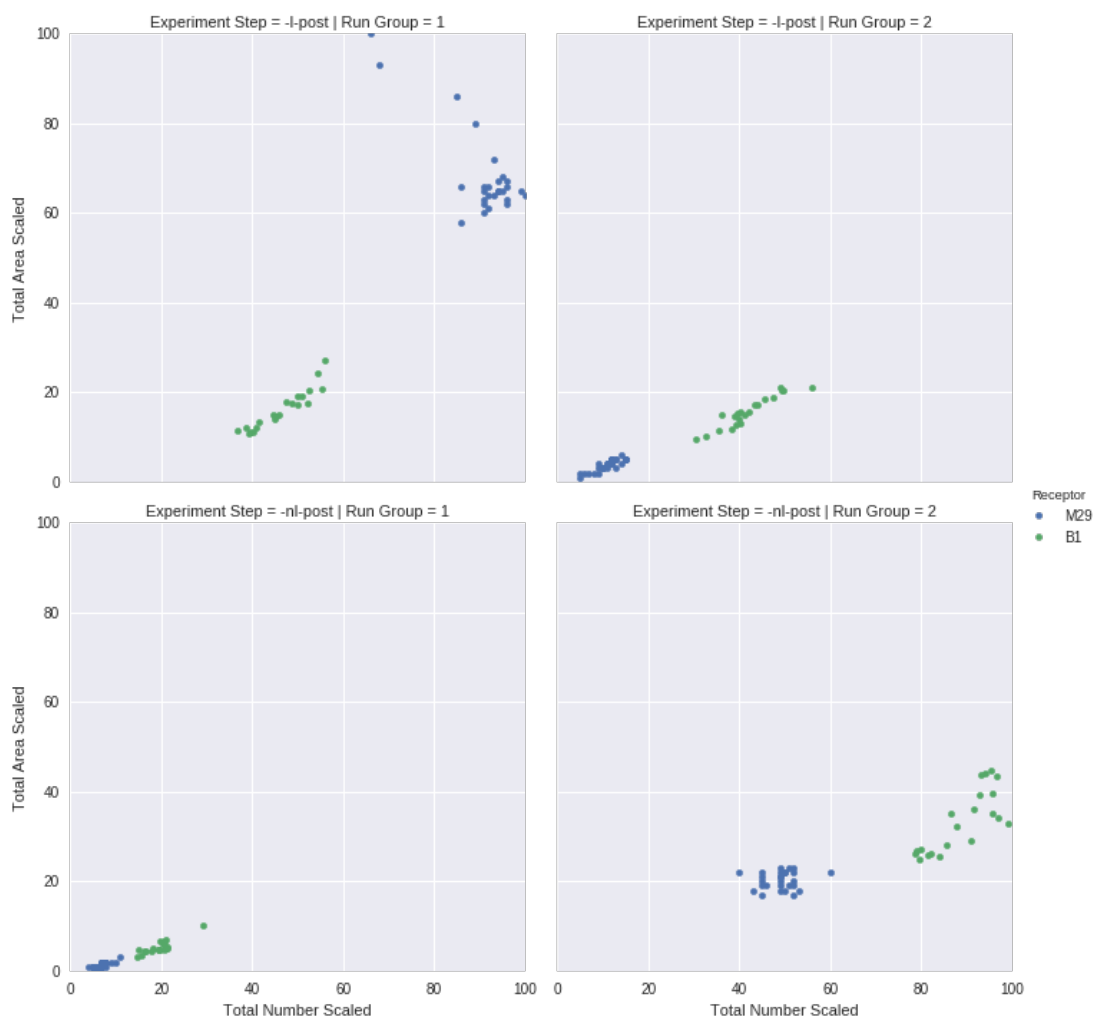
In [7]: post = ['-l-post', '-nl-post']
receptors = ['B1', 'M29']

tmpdf = None
tmpdf = pd.DataFrame(combined65[(combined65['Receptor'].isin(receptors)) &
                                (combined65['Experiment Step'].isin(post))
                                ])

g = sns.FacetGrid(tmpdf,
                  col='Run Group',
                  row='Experiment Step',
                  hue='Receptor',
                  size=5,
                  ylim=(0,100), xlim=(0,100),
                  aspect=1
                  )

g.map(plt.scatter, 'Total Number Scaled', 'Total Area Scaled')
g.add_legend()
g.savefig('comb65_num-by-area_b1_post-steps_scatter.png')

```



Number by area, A3, post steps, time plot

Note: The time axis goes to 25 steps.

```

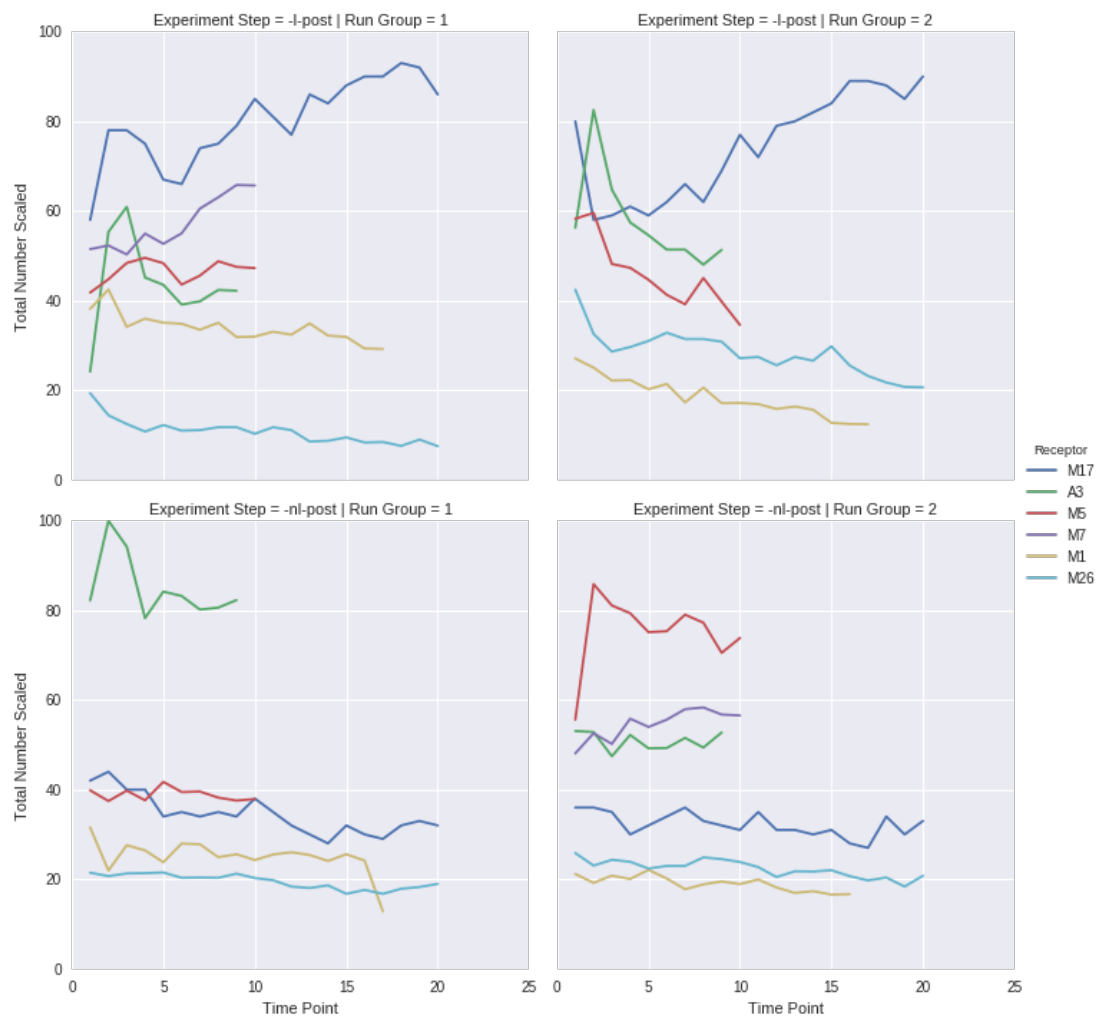
In [8]: post = ['-l-post', '-nl-post']
receptors = ['A3', 'M1', 'M5', 'M7', 'M17', 'M26']

tmpdf = None
tmpdf = pd.DataFrame(combined65[(combined65['Receptor'].isin(receptors)) &
                                (combined65['Experiment Step'].isin(post))
                                ])

g = sns.FacetGrid(tmpdf,
                  col='Run Group',
                  row='Experiment Step',
                  hue='Receptor',
                  size=5,
                  ylim=(0,100), xlim=(0,25),
                  aspect=1
                  )

g.map(plt.plot, 'Time Point', 'Total Number Scaled')
g.add_legend()
g.savefig('comb65_num-by-area_a3_post-steps.png')

```



Number by area, B1, post steps, time plot

Note: The time axis goes to 30 steps.


```

In [9]: post = ['-l-post', '-nl-post']
receptors = ['B1', 'M29']

tmpdf = None
tmpdf = pd.DataFrame(combined65[(combined65['Receptor'].isin(receptors)) &
                                (combined65['Experiment Step'].isin(post))
                                ])

g = sns.FacetGrid(tmpdf,
                  col='Run Group',
                  row='Experiment Step',
                  hue='Receptor',
                  size=5,
                  ylim=(0,100), xlim=(0,30),
                  aspect=1
                  )

g.map(plt.plot, 'Time Point', 'Total Number Scaled')
g.add_legend()
g.savefig('comb65_num-by-area_b1_post-steps.png')

```

